Riya Anil Zawar

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Summary

Computer Engineering graduate specialising in Machine Learning, Deep Learning, and Computer Vision, with experience building end-to-end ML pipelines. Proven track record across research internships (Hiroshima University, DRDO) and a national hackathon win (Smart India Hackathon, Runner-Up among 80,000+ participants).

Education

Cummins College of Engineering for Women **2022 – 2026**
Bachelor of Technology in Computer Engineering (Honors: Data Science & ML); CGPA: 8.86 *Pune, India*

Experience

Visual Information Science Lab, Hiroshima University — Research Intern **Jun 2025 – Feb 2026**
Machine Learning / Computer Vision Research *Hiroshima, Japan*

- Developed a multi-model object detection pipeline on the KITTI dataset, integrating super-resolution preprocessing with YOLOv8 to improve detection of distant objects.
- Benchmarked SR techniques (diffusion models, GANs, OpenCV baseline) for accuracy and compute trade-offs.
- Built the end-to-end pipeline independently as sole contributor within 8 weeks; manuscript in preparation.

ATRC, R&DE(E), DRDO **Jun 2024 – Aug 2024**
Research Intern *Kalas, Pune*

- Identified PCA as the optimal dimensionality-reduction method for hyperspectral classification, achieving 92% SVM accuracy (220 bands to 22 components, 99% variance retained) – a 25-point gain over ICA.
- Benchmarked six DR techniques (PCA, ICA, LDA, NMF, t-SNE, LLE) on the Indian Pines dataset for accuracy, cost, and information retention.

Projects

IntentAnchor – Domain Intent Graph for Conversational Drift Detection | *Python, NLP, MiniLM*

- Co-developed a real-time NLP framework modeling multi-turn dialogue as traversal of a Domain Intent Graph (DIG) to detect semantic intent drift without fine-tuning or context reset.
- Built a turn-level divergence detector (Jensen-Shannon divergence in semantic intent space) achieving 14.41 ms/turn detection latency.
- Built the offline-to-online pipeline (MiniLM classification, Laplace-smoothed transition matrix, drift injection) on a 10K-conversation dataset (22,379 evaluation turns).

VAE Latent Geometry for Dataset Auditing | *Python, PyTorch, Variational Autoencoders, UMAP*

- Co-authored a label-free, model-free dataset auditing framework using VAE latent geometry to detect noisy, mislabeled, and out-of-distribution samples with no ground-truth corruption flags.
- Derived three per-sample audit metrics – KL divergence, Mahalanobis distance, reconstruction probability – achieving AUROC = 1.000 (noise) and 0.931/0.929 (label flips/OOD).

Smart India Hackathon — Terrain Classification | *Python, Machine Learning, Team Lead* **2023**

- Led a 6-member team to National Runner-Up among 80,000+ national participants, building a terrain classification web application for a DRDO problem statement.
- Directed ML model selection, exploratory data analysis, and final presentation end-to-end as team lead in sophomore year, with under two years of prior ML experience.

Technical Skills

Languages: Python, Java, SQL
ML/DL Frameworks: PyTorch, TensorFlow, Keras, scikit-learn - Neural Networks, Generative Models, VAE, Model Training & Evaluation
NLP & Dialogue Systems: Hugging Face Transformers, spaCy, NLTK — Semantic Similarity (Embeddings), Intent Classification, Dialogue State Tracking, Drift Detection (Jensen-Shannon Divergence)
Computer Vision: OpenCV, YOLOv8 — CNNs, Object Detection, Super-Resolution, GANs
Data & Analysis: Pandas, NumPy, Matplotlib, Seaborn, UMAP, Dimensionality Reduction
Tools & Practices: Git, Docker, Data Preprocessing, Feature Engineering, Model Deployment

Leadership & Certifications

TEDx CCOEW **2023-2026**
Co-Head - Content Creation Team *Cummins College of Engineering for Women*

- Authored articles, social media content, and mailer campaigns for the chapter's annual TEDx event.

ACM-W CCEW **2024-2026**
Event Management Team *Cummins College of Engineering for Women*

- Coordinated logistics for chapter technical events and workshops.

Japanese Language Certificate (A+), SPPU